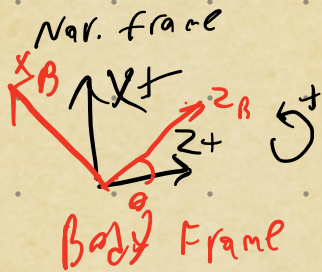


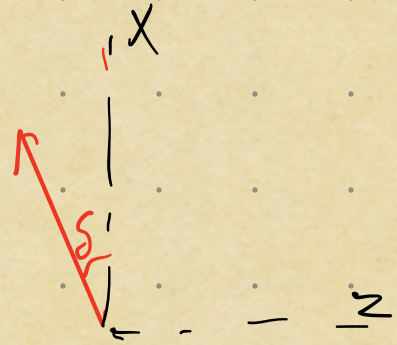
3D of EoM

$$\frac{d\vec{p}_{sys}}{dt} = \sum \vec{F} \quad \frac{d\vec{H}_{sys}}{dt} = \sum \vec{M}$$



model as const.,
will change as
fuel depletes

Body Frame



δ will lead to $-z$ comp.

$$\left(\sum F \right)_N = \vec{T}(t)_N + m \vec{g}_N = R_{NB} \begin{pmatrix} T(t) \cos \delta \\ 0 \\ T(t) \sin \delta \end{pmatrix} - m \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}$$

Passive

$$R_{NB} = R_{BN}^T = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \Rightarrow T(t)_N = \begin{pmatrix} T(t)(\cos \delta \cos \theta + \sin \delta \sin \theta) \\ 0 \\ T(t)(\cos \delta \sin \theta - \sin \delta \cos \theta) \end{pmatrix}$$

$$= T(t) \begin{pmatrix} \cos(\theta - \delta) \\ 0 \\ \sin(\theta - \delta) \end{pmatrix}$$

$$m \ddot{\vec{r}}_N = T(t) \begin{pmatrix} \cos(\theta - \delta) \\ 0 \\ \sin(\theta - \delta) \end{pmatrix} - m \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}$$

$$\ddot{\vec{r}}_N = \frac{T(t)}{m} \begin{pmatrix} \cos(\theta - \delta) \\ 0 \\ \sin(\theta - \delta) \end{pmatrix} - g$$

$$\ddot{x}_N = \frac{T(t)}{m} (\cos(\theta(t)) - \delta(t)) - g$$

$$\ddot{z}_N = \frac{T(t)}{m} (\sin(\theta(t)) - \delta(t))$$

$$(\Sigma^M)_N = (\Sigma^M)_B = \vec{r}_{T+CG} \times \vec{T}(t)_B = \begin{pmatrix} r_{T+CG} \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} T(t) \cos(\delta) \\ 0 \\ -T(t) \sin(\delta) \end{pmatrix} = \begin{pmatrix} 0 \\ r_{T+CG} T(t) \sin(\delta) \\ 0 \end{pmatrix}$$

$$I \ddot{\theta}_B = r T(t) \sin(\delta(t))$$

$$\dot{\chi}_N = \frac{T(t)}{m} (\cos(\delta(t) - \theta(t)) - g) \quad (1)$$

$$\dot{z}_N = \frac{T(t)}{m} (\sin(\delta(t) - \theta(t))) \quad (2)$$

$$\ddot{\theta}_N = \frac{r}{I} T(t) \sin(\delta(t)) \approx \frac{r}{I} T(t) \delta(t) \quad (3)$$

$$\sin(\alpha) \approx \alpha$$



Documentation: I referenced my old Mech 320 notes to make sure I set the problem up right. I used my previous notes on DCMs to find the appropriate DCM for this problem.

